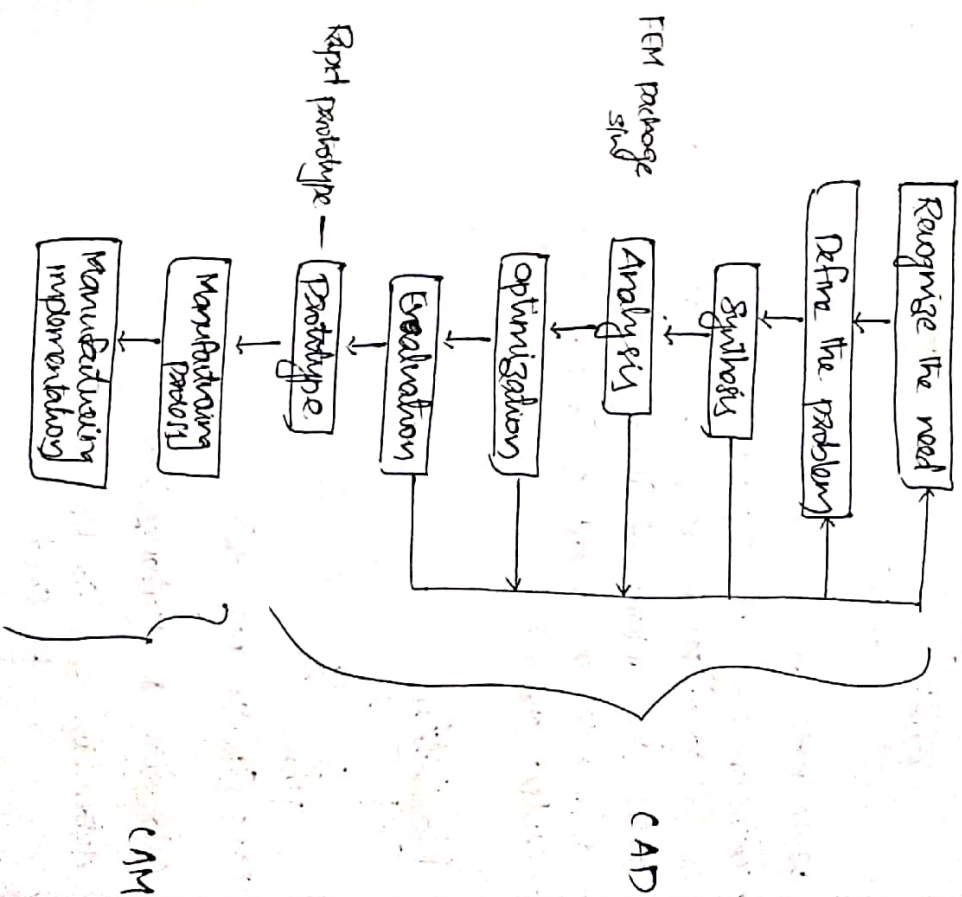
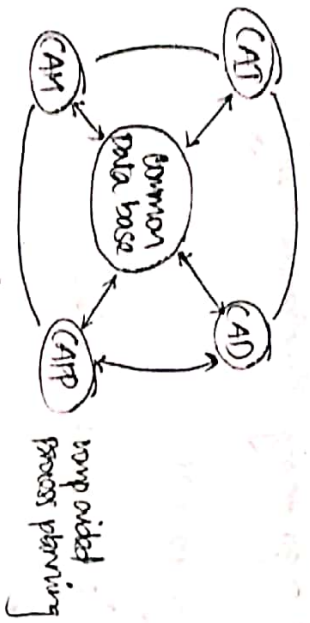


CAD / CAM

Sequential engineering (Traditional engg.) and
Concurrent engineering (Simultaneous engg.)

Engg. Design process

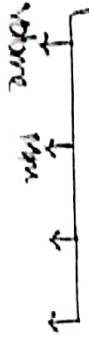




2) waste notes on

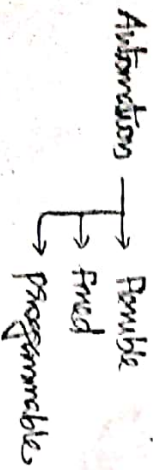
- conversion eng. vs sequential eng.
- role of computers in design as well as manufacturing
- ITC (Int. Transfer of Things)
- short notes on atleast 5 data packages.
- CAD/CAM standards

Flexibility



FMS (Flexible Mfg. Sys)

Automation & Reconfiguration



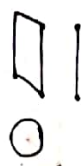
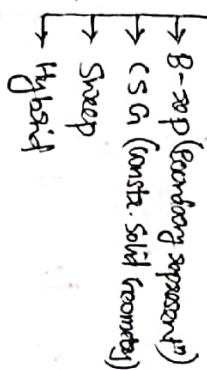
Geometric Modelling

APDL (Ansys Parametric Design Language)
 Work Bench (WB) - Design modeller

space claim (2016)

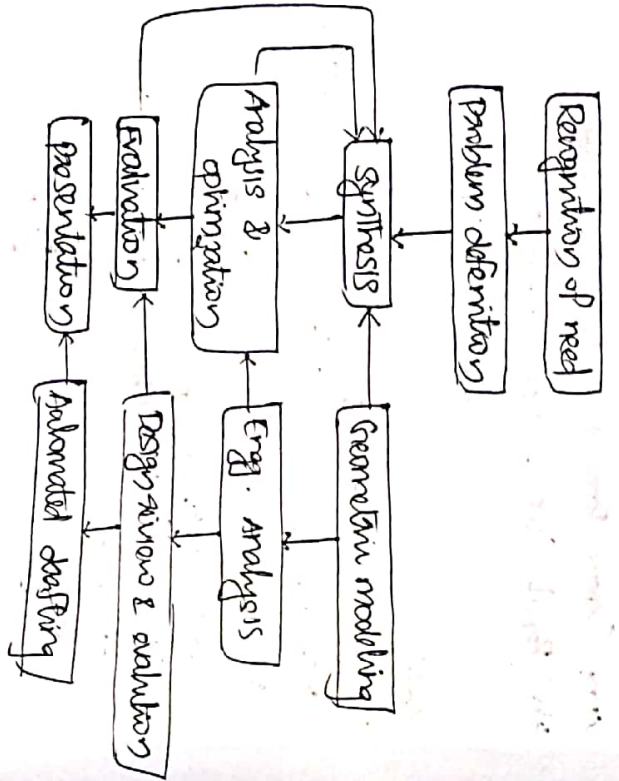
Models

- 1) wire frame model (highly unambiguous, less memory)
- 2) surface model
- 3) solid model

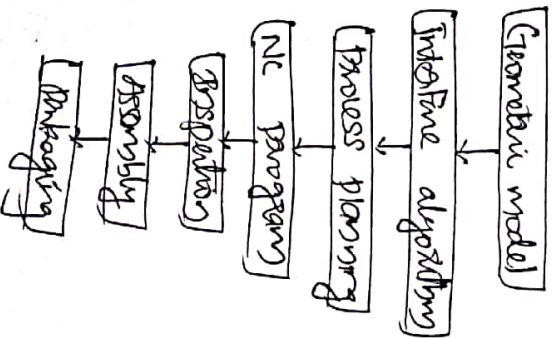


NURBS (Non-Uniform Rational B-splines) representation

The Design Process



CAM Process

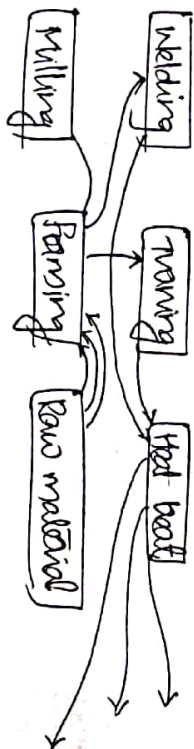


Content Eng

to make decisions consistently
* to make

* Quality Function Deployment (QFD)

Traditional process layout



3 types of layout

- Fixed layout
- product "
- process "

Process layout

Product large & different
Resources general purpose

product layout

small & efficient
specialized

2) no element description

(Jeter)

Must be
Preliminary diagram

- 1 place from on work holder & clamp
- 2 Assemble plug heater to frame
- 3 wire power cord to motor
- 4 wire power cord to switch
- 5

standards

IEEE

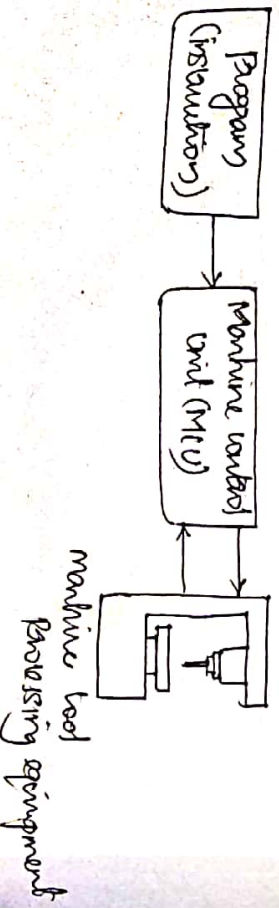
DXF

STEP

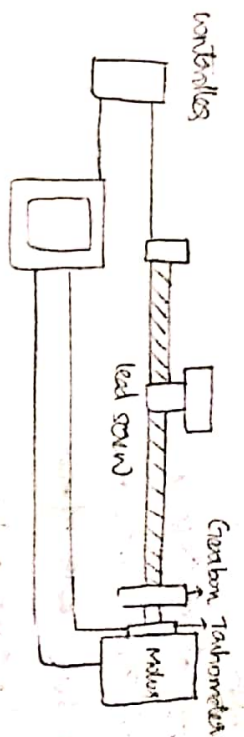
Types of Data

- 1) Nominal
- 2) ordinal
- 3) Interval
- 4) Ratio

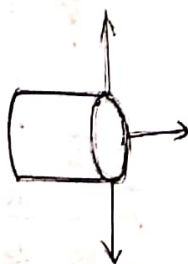
Basic components of an NC system



Major components of an NC machine Tool

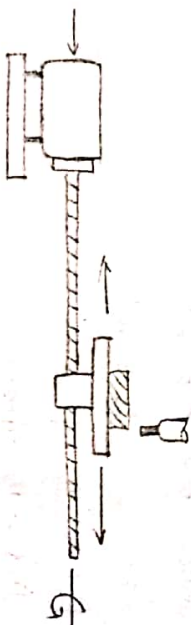


(coordinate axis for rotational work)



Zero point & Target point

Open loop & closed loop



2) There is a grinding m/c in which coolant is to be applied automatically if

- Temperature is $\geq 60^\circ\text{C}$
- Surface speed $\geq 20\text{ m/s}$
- Table feed $\leq 15\text{ mm/min}$

- Temp $\geq 60^\circ$
- Surface speed $< 20\text{ m/s}$
- Table speed $> 15\text{ mm/min}$

- Temp $\geq 60^\circ$
- Surface speed $\geq 20\text{ m/s}$
- Table feed $> 15\text{ mm/min}$

If the sensors are set up in such a way that

A=0 when Temp is $> 60^\circ < 60^\circ$

A=1 when Temp $\geq 60^\circ$

B=0 when Surface speed $< 20\text{ m/s}$

B=1 " $\geq 20\text{ m/s}$

C=0 when feed $\leq 15\text{ mm/min}$

C=1 when feed $> 15\text{ mm/min}$

Make a circuit to implement this

S=1 coolant should be on

S=0 otherwise

A	B	C	S ₁
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	1

$$S_{\text{cool}} = A.B.C + A.B.C' + A.B'.C$$

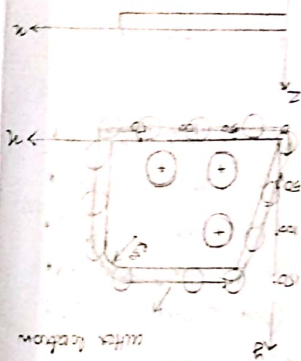
Classification of NC

Based on the following:

- 1) Motion control PTP-CP (Point-to-point - continuous path)
- 2) control loop - open loop & closed loop
- 3) Power driven - hydraulic, electric, pneumatic
- 4) Positioning system - incremental & absolute
- 5) Hardwired NC & software NC

N07 G01 X35
 N08 G01 X0Y0
 N09 G40 G00 X-40 Y-50
 N10 G00 X0 Y-50.0
 N01 M30

CNC programming Basics
 G code - preparatory code
 M - miscellaneous code
 G00 - Rapid traverse
 G01 - Linear interpolation
 G02 - circular interpolation clockwise
 G03 - circular interpolation counter clockwise
 2 axis milling
 Tools : $\phi 20$ mm end mill with 4 teeth
 Parameters
 Speed (S) : 1000 rev/min
 Feed (F) : 50 mm/min
 Machine tool planning
 First program
 Simulation
 Let the tool is at $x=0, y=0, z=110$
 N01 G01 G02 X0 Y-0.5 Z0 (Left side of oval)
 N02 G00 Z-50 M03 (Feed to start depth) turn spindle on
 N03 G01 G02 X0.5 Y0.5 (Edge part - start with offset
 (with straight edge)
 N04 G01 X150
 N05 G01 Y0.0
 N06 G01 X0 Y0.5
 N07 G01 X0 Y0.5 (could not finish - cut)



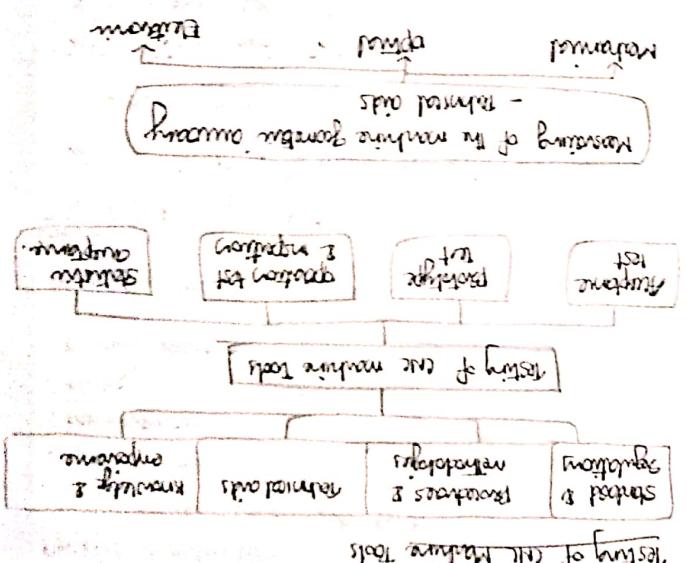
- Accuracy
- Precision
- Sensitivity
- Consistency
- Resolution
- Threshold

- Drift
- Stability
- Tolerance
- Long span

The set of values defined for the instrument which are used to measure the quantity with out disturbing it.

Static Uncertainty

Hysteresis & Backlash



$$\begin{bmatrix} g' \\ n' \end{bmatrix} = \begin{bmatrix} g \\ n \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & s_n \end{bmatrix}$$

$$\begin{bmatrix} g' \\ n' \end{bmatrix} = \begin{bmatrix} g \\ n \end{bmatrix} + \begin{bmatrix} dg \\ dn \end{bmatrix}$$

$$n' = n + dn$$

$$g' = g + dg$$

Translation

$$\begin{bmatrix} g' \\ n' \end{bmatrix} = \begin{bmatrix} g \\ n \end{bmatrix} + \begin{bmatrix} g_{\text{rot}} - g_{\text{rot}} \\ n_{\text{rot}} - n_{\text{rot}} \end{bmatrix}$$

$$n' = n_{\text{rot}} - g_{\text{rot}}$$

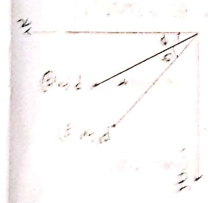
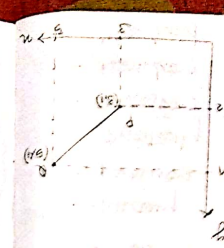
$$n' = n_{\text{rot}} (g + g') - n (g_{\text{rot}} - g_{\text{rot}})$$

$$n' = n_{\text{rot}} (g + g')$$

(Rotation Transformation)

- Translation
- Scaling
- Rotation
- Shear

Rotation Transformation



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Its coordinates in F_0 are

$$p = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Consider a point defined in F_0 by

$$p = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}$$

To rotate by θ , rotate to F_1 and rotate back.

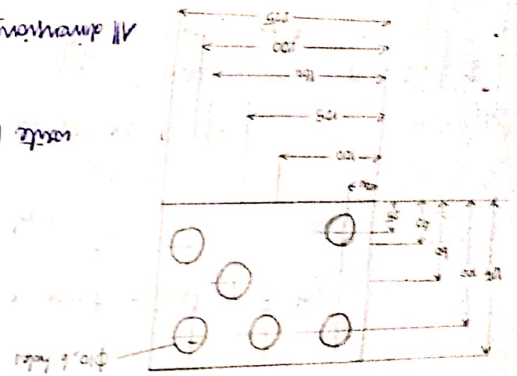
$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

- control - RTT, Autonomous, intelligent
- rigid or flexible
- region of freedom
- fixed or mobile
- shape of work envelope
- power source

Classification of robot based on

3) Discuss the classification of robots

write the RTT program for the same
All dimensions are in mm



origin in the x-y-z coord sys.
of origin. Use the lower left corner of the part as the
the part is 12 mm thick cutting speed = 100 m/min, and feed =
2) a) write a program to drill holes in the part shown in fig

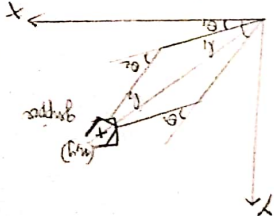
- 1) robot resolution
- 2) its position
- 3) robot arm joint

1) Discuss

Robot programming

and effector motion

calculate the joint motion required to generate a desired



Inverse kinematics

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

Forward kinematics

Pose - Position and orientation

- 9) Apps of robots
- 1) robotic sensor
 - 2) vision
 - 3) range detector
 - 4) force or torque